LESSON 01 / 07

The problem: no trig function is one-to-one

A function has an inverse only if it is one-to-one (injective). Every trig function fails globally: $\sin 0 = \sin \pi = 0$, $\tan 0 = \tan \pi = 0$, and so on. Graphically, a horizontal line $y = c$ with $|c| < 1$ crosses $y = \sin x$ infinitely many times. So "$\sin^{-1}$" cannot exist on all of $\mathbb{R}$ until we **cut the domain down to a strip on which sine is one-to-one and takes every value in** $[-1, 1]$.

> INTUITION
> **Inverse = undo: the one question that decides everything**
> $y = \sin^{-1} x$ means exactly: "y is the angle (in the chosen strip) whose sine is x". So $\sin(\sin^{-1} x) = x$ is automatic on the domain of $\sin^{-1}$ — but $\sin^{-1}(\sin y) = y$ only for y inside the chosen strip. **All the difficulty of this chapter lives on that one asymmetry.** The strip is a convention — but once fixed (by the official principal-value ranges below), every formula is determined, and every "simplify" question is a branch-check question.

The six principal values (memorise the ranges — they are the whole syllabus)

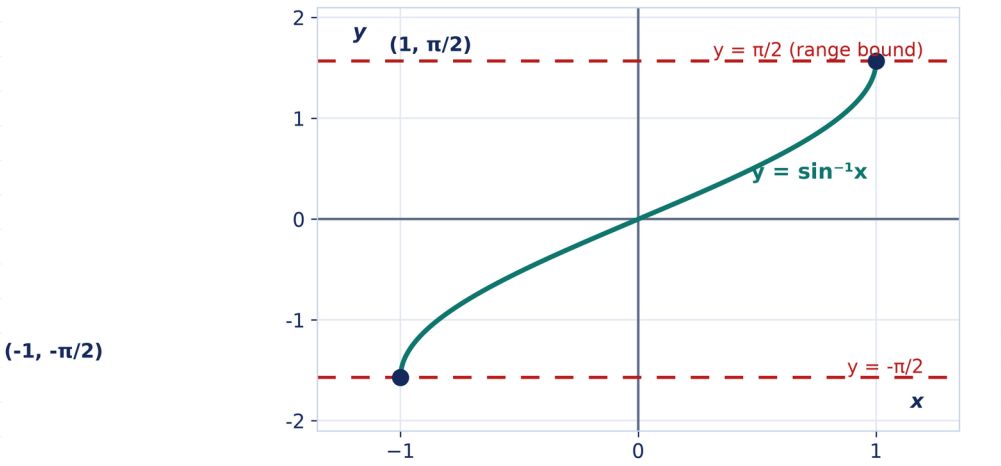
Function	Domain	Principal range (the branch)
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$\lvert x\rvert \geq 1$	$[0, \pi] \setminus \{\pi/2\}$
$\csc^{-1} x$	$\lvert x\rvert \geq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$

Reading the table correctly: $\sin^{-1}$ is the inverse of the restriction $\sin : [-\pi/2, \pi/2] \to [-1, 1]$; $\cos^{-1}$ inverts $\cos : [0, \pi] \to [-1, 1]$; $\tan^{-1}$ inverts $\tan : (-\pi/2, \pi/2) \to \mathbb{R}$. The half-open endpoints for $\tan^{-1}$ reflect the vertical asymptotes of $\tan$.

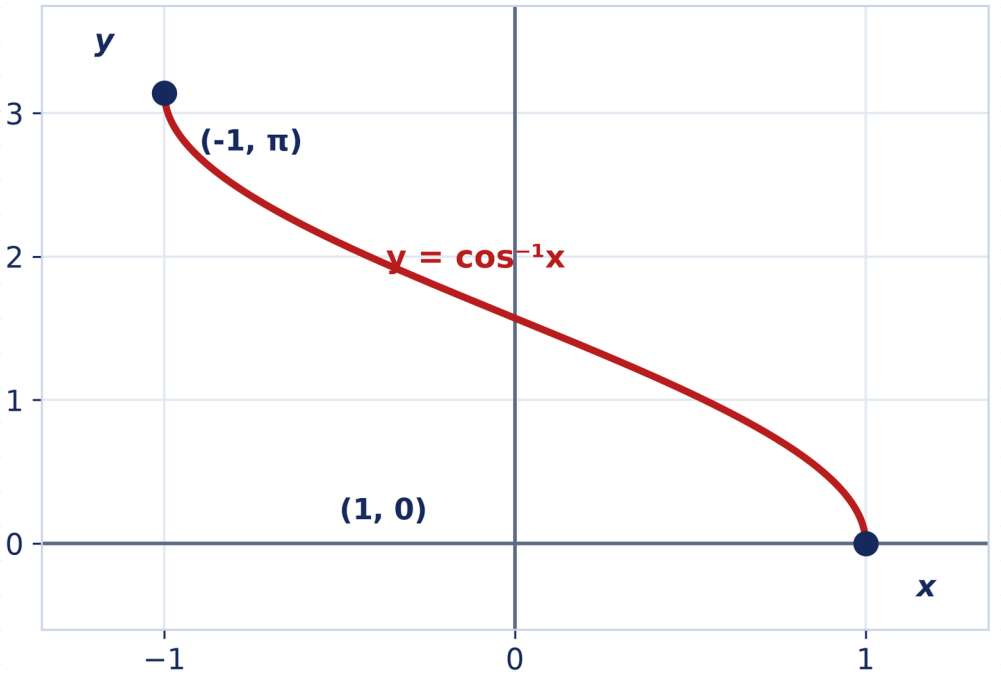
> ASSUMPTIONS & CONVENTIONS
> **Symbols and conventions used everywhere below**
> - $\sin^{-1} x$ means the **inverse function** (arcsine), never $\csc x = (\sin x)^{-1}$. Write $\sin^{-1}$ / arcsin for the inverse; never write $\sin^{-1} x$ for $1/\sin x$.
> - All inputs x are real; ranges above are in **radians**. Degree-mode answers convert at the boundary only.
> - "$\sin^{-1}(\sin y) = y$"-type claims always silently assume y lies in the principal range of the outer inverse. Part of the answer is stating that assumption.
> - $\cot^{-1}$ takes range $(0, \pi)$ (the JEE convention); some calculus texts use $(-\pi/2, 0) \cup (0, \pi/2]$ — the value of $\cot^{-1} x + \tan^{-1} x$ then changes. We fix $(0, \pi)$ throughout and note the caveat here once.

Graphs: domain on the x-axis, branch on the y-axis

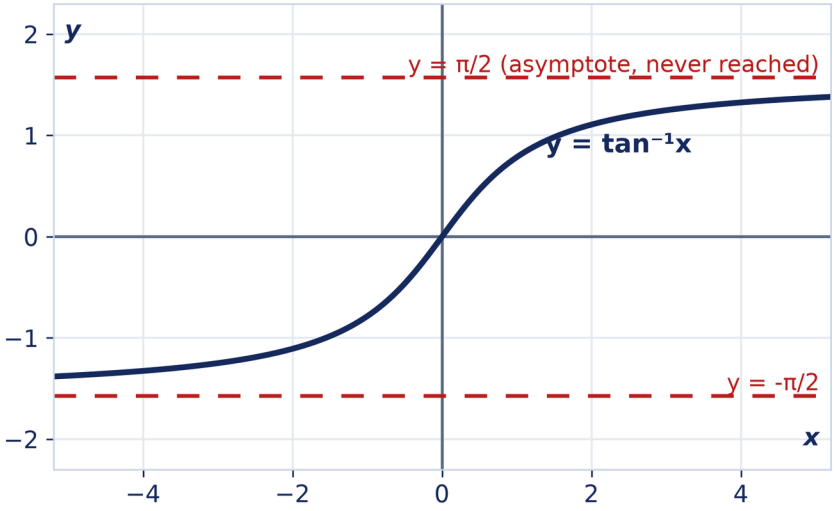
y = sin⁻¹x : domain [-1, 1] ⊂ x-axis, branch [-π/2, π/2] ⊂ y-axis



y = cos⁻¹x : range [0, π] — note cos⁻¹1 = 0, cos⁻¹(-1) = π



y = tan⁻¹x : range (-π/2, π/2) is OPEN — the asymptotes are not values



*Three graph-reading habits: (i) the **branch bounds are horizontal lines on the y-axis** — they constrain the output; (ii) endpoints may be attained ($\sin^{-1}$, $\cos^{-1}$) or not ($\tan^{-1}$); (iii) $\sin^{-1}$ and $\tan^{-1}$ are **odd** functions ($\sin^{-1}(-x) = -\sin^{-1}x$, $\tan^{-1}(-x) = -\tan^{-1}x$), visible as origin-symmetry; $\cos^{-1}$ is **neither** — its graph is not symmetric.*

The three complementary-pair identities (state + prove)

KEY RESULT

Complementary sums — valid on the stated domains

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \qquad |x| \le 1$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \qquad x \in \mathbb{R}$$

$$\sec^{-1}x + \csc^{-1}x = \frac{\pi}{2}, \qquad |x| \ge 1$$

SHORT DERIVATION

Proof of the first — the template for all three

Let $\theta = \sin^{-1}x$, so $\theta \in [-\pi/2, \pi/2]$ and $\sin\theta = x$. Then $\cos(\pi/2 - \theta) = \sin\theta = x$, and $\pi/2 - \theta \in [0, \pi]$ — which is exactly the principal range of $\cos^{-1}$. Hence $\cos^{-1}x = \pi/2 - \theta = \pi/2 - \sin^{-1}x$. ∎

The second: $\cot(\pi/2 - \theta) = \tan\theta$, and the range bookkeeping ($\pi/2 - \tan^{-1}x \in (0, \pi)$) is identical. Same for the third. **The mechanism is always: co-function identity + range verification.** The range check is not decoration — it is what makes the inverse principal.

Standard values (the x-values worth instant recall)

x	$\sin^{-1}x$	x	$\cos^{-1}x$	x	$\tan^{-1}x$
0	0	0	$\pi/2$	0	0
$1/2$	$\pi/6$	$1/2$	$\pi/3$	$1/\sqrt{3}$	$\pi/6$
$1/\sqrt{2}$	$\pi/4$	$1/\sqrt{2}$	$\pi/4$	1	$\pi/4$
$\sqrt{3}/2$	$\pi/3$	$\sqrt{3}/2$	$\pi/6$	$\sqrt{3}$	$\pi/3$
1	$\pi/2$	1	0	—	$\to \pi/2$
$-1/2$	$-\pi/6$	$-1/2$	$2\pi/3$	-1	$-\pi/4$
-1	$-\pi/2$	-1	π	—	$\to -\pi/2$

Note the systematic pattern: $\sin^{-1}$ *and* $\cos^{-1}$ *of the same* x *always sum to* $\pi/2$*;* $\tan^{-1}$ *of the reciprocal of* $\sqrt{3}$*-values is the complementary angle. Negative inputs:* $\sin^{-1}$ *and* $\tan^{-1}$ *give negative outputs (odd functions);* $\cos^{-1}(-x) = \pi - \cos^{-1}x$ *(output stays in* $[0, \pi]$*).*

COMMON TRAP

Common traps — foundations

- $\sin^{-1}(\sin\theta) = \theta$ *is* ***false*** *for e.g.* $\theta = \frac{3\pi}{4}$*: the left side is* $\pi/4$*. Trap of the entire chapter.*
- $\sin(\sin^{-1}x) = x$ *holds for all* $x \in [-1, 1]$*;* $\sin^{-1}(\sin x) = x$ *holds only for* $x \in [-\pi/2, \pi/2]$*.*
- *Domain before value:* $\sin^{-1}2$ *is* ***undefined*** *in* $\mathbb{R}$*, not "*$\pi/2$*".* $\cos^{-1}(1.01)$ *does not exist.*
- $\tan^{-1}x$ *never equals* $\pm\pi/2$*; writing* $\tan^{-1}x = \pi/2$ *"as* $x \to \infty$*" is a limit, not a value.*
- $(\sin x)^{-1} = \csc x \neq \sin^{-1}x$*. The notation is genuinely ambiguous in print; our convention is stated in the ASSUMPTIONS box and used without exception.*
- $\cos^{-1}x$ *is not an odd function:* $\cos^{-1}(-1/2) = 2\pi/3 \neq -\pi/3$*.*

LESSON 02 / 07

The one skill: reduce the angle into the principal branch

For every "inverse of a trig" the question is: *which angle in the principal range has the same trig value as x?* The answer is never "x itself" unless x already lies in the branch. The reliable method is a **two-step reduction** (shown for $\sin^{-1}(\sin x)$; the others are identical in spirit):

KEY RESULT

Reduction algorithm for $f^{-1}(fx)$ — reduce, then read the branch

1. **Reduce mod the period** of f: replace x by $t = x - 2k\pi$ (sine/cosine) or $t = x - k\pi$ (tangent/cotangent), with $k \in \mathbb{Z}$ chosen so that t falls in one full period.
2. **Read the branch**: find the unique angle y in the principal range with $f(y) = f(t)$; that y is the answer.

$\sin^{-1}(\sin x)$ in closed piecewise form

Reduce x to $t \in [-\pi/2, 3\pi/2)$ by subtracting $2\pi k$. Then the closed piecewise read is just two cases:

$$\sin^{-1}(\sin t) = t \quad \text{for } t \in [-\pi/2, \pi/2]$$

$$\sin^{-1}(\sin t) = \pi - t \quad \text{for } t \in (\pi/2, 3\pi/2)$$

Why $\pi - t$: for $t \in (\pi/2, 3\pi/2)$, the angle $\pi - t \in (-\pi/2, \pi/2)$ and $\sin(\pi - t) = \sin t$ — it is the mirror angle on the branch, produced by the symmetry of the sine wave about $t = \pi/2$.

WORKED EXAMPLE

Worked — $\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$ and $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)$

$\frac{7\pi}{6} = \pi + \frac{\pi}{6}$ is in $(\pi/2, 3\pi/2)$: answer $= \pi - \frac{7\pi}{6} = -\frac{\pi}{6}$. Check: $\sin\frac{7\pi}{6} = -\frac{1}{2} = \sin(-\frac{\pi}{6})$ ✓, and $-\frac{\pi}{6}$ is on the branch ✓.

$\frac{5\pi}{4} \in (\pi/2, 3\pi/2)$: answer $= \pi - \frac{5\pi}{4} = -\frac{\pi}{4}$. Check: $\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2} = \sin(-\frac{\pi}{4})$ ✓.

For $\sin^{-1}\left(\sin\frac{9\pi}{5}\right)$: first reduce — $\frac{9\pi}{5} - 2\pi = -\frac{\pi}{5} \in [-\pi/2, \pi/2]$, so the answer is $-\frac{\pi}{5}$.

Reduce first, read second — skipping the reduction is how wrong signs appear.

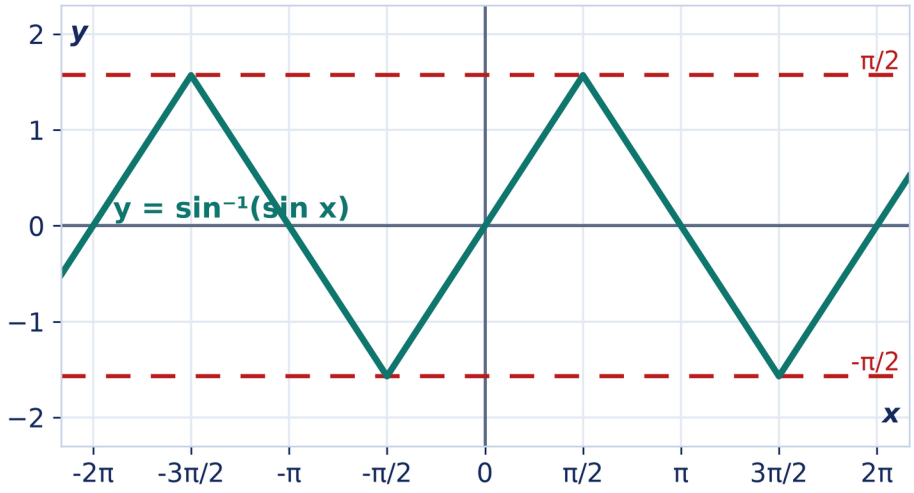
$\cos^{-1}(\cos x)$

Reduce to $t \in [0, 2\pi)$; then $\cos^{-1}(\cos t) = t$ for $t \in [0, \pi]$ and $= 2\pi - t$ for $t \in (\pi, 2\pi)$ (mirror about $t = 0$ on the circle: $2\pi - t \in (0, \pi)$ has the same cosine). E.g. $\cos^{-1}(\cos\frac{7\pi}{4}) = 2\pi - \frac{7\pi}{4} = \frac{\pi}{4}$.

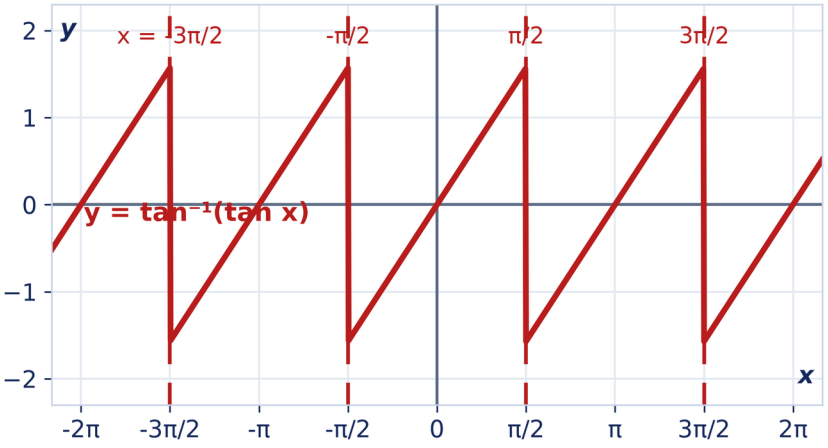
$\tan^{-1}(\tan x)$

Reduce to $t \in (-\pi/2, \pi/2)$ by subtracting πk (period π, not 2π!): $\tan^{-1}(\tan x) = x - \pi k$ with the unique $k \in \mathbb{Z}$ landing the result in $(-\pi/2, \pi/2)$. E.g. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right) = \frac{3\pi}{4} - \pi = -\frac{\pi}{4}$, $\tan^{-1}(\tan 5) = 5 - 2\pi$ (since $5 - 2\pi \approx -1.28 \in (-\pi/2, \pi/2)$). Same for $\cot^{-1}(\cot x)$ with output forced into $(0, \pi)$.

sin⁻¹(sin x) : a sawtooth oscillating INSIDE the branch [-π/2, π/2]



tan⁻¹(tan x) : period π, jumps at odd multiples of π/2 (undefined there)



The reverse compositions: trig of an inverse

$\sin(\tan^{-1} x)$, $\tan(\cos^{-1} x)$ and friends never need formulas — draw the **reference right triangle**:

STANDARD APPLICATION

The triangle method — every trig-of-inverse in one picture

Given $\theta = \tan^{-1} x$: θ is the acute-or-negative angle with $\tan\theta = x = \frac{x}{1}$. Draw a right triangle with opposite $= x$, adjacent $= 1$, hypotenuse $= \sqrt{1+x^2}$ (the sign of x fixes the quadrant; the triangle gives magnitudes). Read off:

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}, \qquad \cos(\tan^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

valid for all real x (both sides odd/even as they must be). Likewise with $\theta = \sin^{-1} x$: opposite $= x$, hypotenuse $= 1$, adjacent $= \sqrt{1-x^2}$ (non-negative because $\theta \in [-\pi/2, \pi/2]$, where $\cos\theta \geq 0$):

$$\cos(\sin^{-1} x) = \sqrt{1-x^2},$$

$$\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}, \quad |x| < 1$$

For $\theta = \cos^{-1} x$ ($\theta \in [0, \pi]$): adjacent $= x$, hypotenuse $= 1$, opposite $= \sqrt{1-x^2}$ (again $\sin\theta \geq 0$ on the branch):

$$\sin(\cos^{-1} x) = \sqrt{1-x^2},$$

$$\tan(\cos^{-1} x) = \frac{\sqrt{1-x^2}}{x}, \quad x \neq 0$$

The **sign of the square root** is decided by the branch, not by taste: $\sqrt{1-x^2}$ appears with a $+$ because both $\cos(\sin^{-1} x)$ and $\sin(\cos^{-1} x)$ are non-negative on their branches. If you ever need $\cos(\sin^{-1} x)$ for an angle outside a branch — you don't, that's the point of branches.

SHORT DERIVATION

Worked — simplify $\tan(\sin^{-1} x)$ for $x \in (-1, 1)$ and evaluate $\sin\left(\cos^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13}\right)$

First part: from the triangle, $\frac{x}{\sqrt{1-x^2}}$.

Second: let $A = \cos^{-1}\frac{3}{5} \in [0, \pi]$, $B = \sin^{-1}\frac{5}{13} \in [-\pi/2, \pi/2]$. Then $\sin A = \frac{4}{5}$ (positive on the branch!), $\cos B = \frac{12}{13}$ (positive on the branch!). So

$$\sin(A+B)$$

$$= \sin A\cos B + \cos A\sin B$$

$$= \frac{4}{5}\cdot\frac{12}{13} + \frac{3}{5}\cdot\frac{5}{13}$$

$$= \frac{48+15}{65} = \frac{63}{65}.$$

The trap: $\sin A = \pm 4/5$ in general; the branch of $\cos^{-1}$ forces $+$.

COMMON TRAP

Common traps — branch reading

- $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \frac{\pi}{4}$, **not** $\frac{3\pi}{4}$ — $\frac{3\pi}{4}$ is outside $[-\pi/2, \pi/2]$.
- $\tan^{-1}(\tan x)$ has period π and is undefined at $x = \frac{\pi}{2} + k\pi$; $\sin^{-1}(\sin x)$ has period 2π.
- $\cos(\sin^{-1} x) = +\sqrt{1-x^2}$ always (branch $\Rightarrow$ cosine ≥ 0); never $\pm$.
- $\tan(\cos^{-1} x)$ is positive for $x > 0$ and negative for $x < 0$ — the triangle gives magnitudes, the branch gives signs.
- Reducing *before* reading: $\sin^{-1}\left(\sin\frac{9\pi}{5}\right) = -\frac{\pi}{5}$ requires subtracting 2π first; treating $\frac{9\pi}{5}$ as "in $(\pi/2, 3\pi/2)$" is wrong because the two-case read assumes the reduction already happened.

LESSON 03 / 07

The master formula and why conditions are unavoidable

Take tangents of $\tan^{-1}x \pm \tan^{-1}y$ using the compound-angle formula. The tangent of the answer is never the issue — the issue is which **branch of** $\tan^{-1}$ the sum lands in. Since each of $\tan^{-1}x, \tan^{-1}y \in (-\pi/2, \pi/2)$, their sum lies in $(-\pi, \pi)$; and $\tan^{-1}$ returns values only in $(-\pi/2, \pi/2)$. Whenever the true sum exits that window, a $\pm\pi$ correction is mandatory.

KEY RESULT

Addition and subtraction — statement with corrections

For all real x, y:

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\frac{x+y}{1-xy} \quad \text{if } xy < 1$$

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\frac{x+y}{1-xy}$$

$$\text{if } xy > 1 \text{ and } x > 0$$

$$\tan^{-1}x + \tan^{-1}y = -\pi + \tan^{-1}\frac{x+y}{1-xy}$$

$$\text{if } xy > 1 \text{ and } x < 0$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy} \quad \text{if } xy > -1$$

with the analogous $\pm\pi$ corrections when $xy < -1$ (sign chosen so the final answer lies in $(-\pi, \pi)$, matching the true sum).

Domain wall: the fraction is undefined at $xy = 1$ (sum $= \pm\pi/2$) and $xy = -1$ (difference $= \pm\pi/2$) — those are boundary cases, handled by direct branch arithmetic, not by the formula.

SHORT DERIVATION

Derivation of the corrections — sign of the cosine decides

Let $A=\tan^{-1}x$, $B=\tan^{-1}y$, $S=A+B\in(-\pi,\pi)$. First, $\tan S=\frac{x+y}{1-xy}$ (compound angle), so $S=\tan^{-1}\frac{x+y}{1-xy}+k\pi$ for some integer $k\in\{-1,0,1\}$. Which k? Ask where S sits on $(-\pi,\pi)$ via its cosine:

$$\cos S=\cos A\cos B\,(1-\tan A\tan B)$$

$$\cos A\cos B\,(1-xy)$$

and $\cos A,\cos B>0$ strictly (both branches inside $(-\pi/2,\pi/2)$), so $\operatorname{sign}(\cos S)=\operatorname{sign}(1-xy)$:

- $xy<1$: $\cos S>0$ forces $S\in(-\pi/2,\pi/2)$ — the branch itself: $k=0$.
- $xy>1$, $x>0$ (then $y>0$): $S\in(\pi/2,\pi)$: $k=+1$.
- $xy>1$, $x<0$ (then $y<0$): $S\in(-\pi,-\pi/2)$: $k=-1$.
- $xy=1$: $\cos S=0$ and $S=\pm\pi/2$ — compute directly, the fraction is void.

■ (Every line of the table is machine-checked at adversarial points in `math_checks.py`.)

WORKED EXAMPLE

Worked — $\tan^{-1}2+\tan^{-1}3$ and the sign trap

$x=2,y=3$: $xy=6>1$ and $x>0$, so the correction fires:

$$\tan^{-1}2+\tan^{-1}3=\pi+\tan^{-1}\tfrac{5}{1-6}=\pi+\tan^{-1}(-1)$$

$$\pi-\tfrac{\pi}{4}=\tfrac{3\pi}{4}.$$

Without the correction you would answer $-\pi/4$ — wrong quadrant. Sanity: $\tan^{-1}2\approx1.107$, $\tan^{-1}3\approx1.249$, sum $\approx2.356=\frac{3\pi}{4}$ ✓.

By contrast $\tan^{-1}\frac12+\tan^{-1}\frac13$: $xy=\frac16<1$, no correction:

$$\tan^{-1}\tfrac12+\tan^{-1}\tfrac13$$

$$=\tan^{-1}\left(\frac{\frac12+\frac13}{1-\frac16}\right)$$

$$=\tan^{-1}1=\tfrac{\pi}{4}.$$

The 2tan⁻¹x chain — three faces, three conditions

Set $y=x$ in the addition formula, and separately take sin/cos of $2\tan^{-1}x$ via the triangle of sheet 02. All three forms are JEE staples:

KEY RESULT

$2\tan^{-1}x$ *in three forms (conditions are part of the formula)*

$$2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2} \quad |x|<1$$

$$2\tan^{-1}x = \pi + \tan^{-1}\frac{2x}{1-x^2} \quad x>1;$$

$$= -\pi + \tan^{-1}\frac{2x}{1-x^2} \quad x<-1$$

$$2\tan^{-1}x = \sin^{-1}\frac{2x}{1+x^2} \quad |x|\le 1$$

$$2\tan^{-1}x = \pi - \sin^{-1}\frac{2x}{1+x^2} \quad x>1;$$

$$= -\pi - \sin^{-1}\frac{2x}{1+x^2} \quad x<-1$$

$$2\tan^{-1}x = \cos^{-1}\frac{1-x^2}{1+x^2} \quad x\ge 0;$$

$$= -\cos^{-1}\frac{1-x^2}{1+x^2} \quad x<0$$

SHORT DERIVATION

Where the $\sin^{-1}$ ***face comes from***

Let $\theta=\tan^{-1}x$, *so* $\tan\theta=x$. *Then* $\sin 2\theta=\frac{2\tan\theta}{1+\tan^2\theta}=\frac{2x}{1+x^2}$. *Now branch-check: for* $|x|\le 1$, $2\theta\in[-\pi/2,\pi/2]$ *and* $\sin^{-1}(\sin 2\theta)=2\theta$ ✓. *For* $x>1$, $2\theta\in(\pi/2,\pi)$, *so* $2\theta=\pi-\sin^{-1}(\sin 2\theta)$ ✓ *(mirror-angle read of sheet 02). The* $\cos^{-1}$ *face:* $\cos 2\theta=\frac{1-x^2}{1+x^2}$, *and* $2\theta\in[0,\pi]$ *exactly when* $\theta\ge 0$, *i.e.* $x\ge 0$ ✓. *Same computation, three branch bookkeepings.*

STANDARD APPLICATION

Standard applications — simplification and telescoping

1. ***Simplify a sum of two arctans***: *check* xy *against* 1 *first, then apply the table. E.g.* $\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}$: $xy=\frac{1}{56}<1 \to \tan^{-1}\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{56}}=\tan^{-1}\frac{15/56}{55/56}=\tan^{-1}\frac{3}{11}$.

1. ***Sums that telescope***: *recognise* $\tan^{-1}\frac{a-b}{1+ab}=\tan^{-1}a-\tan^{-1}b$ *(when* $ab>-1$*). E.g.*

$$\sum_{k=1}^{n}\tan^{-1}\frac{1}{1+k+k^2}=\sum_{k=1}^{n}[\tan^{-1}(k+1)-\tan^{-1}k]$$

$$\tan^{-1}(n+1)-\frac{\pi}{4}.$$

because $1+k+k^2=1+k(k+1)$ *and* $\frac{(k+1)-k}{1+k(k+1)}=\frac{1}{1+k+k^2}$ ✓. *(Olympiad sheets push this hard; the exam-visible core is the identity itself.)*

1. ***Equations***: $2\tan^{-1}x=\sin^{-1}x$*-style problems: convert both sides to one inverse via the chain, then use* ***injectivity of the outer inverse****, never of* $x\mapsto$ *expression.*

COMMON TRAP

Common traps — the addition formulas

- Forgetting the $\pm\pi$: $\tan^{-1}2+\tan^{-1}3=\frac{3\pi}{4}$, not $-\frac{\pi}{4}$. Rule of thumb: two positive arctans of numbers >1 each... more precisely if $x,y>0$ and $xy>1$ the sum exceeds $\frac{\pi}{2}$ and needs $+\pi$.
- Writing $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$ with **no condition at all** — the identity is false at e.g. $x=y=2$ as written.
- $xy=1$ and $xy=-1$ are boundary cases: the fraction $\frac{x+y}{1-xy}$ explodes while the true sum is $\pm\frac{\pi}{2}$ or $\pm\frac{\pi}{2}\mp\pi$ — compute from the branch directly: if $x=y=1$, $\tan^{-1}1+\tan^{-1}1=\frac{\pi}{2}$ ✓.
- $2\tan^{-1}x=\sin^{-1}\frac{2x}{1+x^2}$ fails for $x>1$: e.g. $x=2$: LHS $=2\tan^{-1}2\approx2.214$, but $\sin^{-1}\frac{4}{5}\approx0.927$ — the $\pi-$ face is the true one ($\pi-0.927\approx2.214$ ✓).
- In telescoping, check $ab>-1$ before using $\tan^{-1}\frac{a-b}{1+ab}=\tan^{-1}a-\tan^{-1}b$; for positive a,b it always holds.

LESSON 04 / 07

$\sin^{-1}x \pm \sin^{-1}y$ — derived, not recited

Let $A=\sin^{-1}x$, $B=\sin^{-1}y$, so $A,B\in[-\pi/2,\pi/2]$ and $A+B\in[-\pi,\pi]$. Take the sine of the sum: $\sin(A+B)=x\sqrt{1-y^2}+y\sqrt{1-x^2}=:\sigma$ (the square roots are non-negative precisely because of the branches). Then $A+B=\sin^{-1}\sigma$ **if** $A+B\in[-\pi/2,\pi/2]$, else the mirror-angle correction applies. The branch test reduces to the sign of

$$\cos(A+B)=\sqrt{1-x^2}\sqrt{1-y^2}-xy$$

which is ≥ 0 exactly when $xy\leq 0$ or $x^2+y^2\leq 1$.

KEY RESULT

The sum formulas, with the branch conditions

$$\sin^{-1}x+\sin^{-1}y=\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } xy\leq 0 \text{ or } x^2+y^2\leq 1$$

$$\sin^{-1}x+\sin^{-1}y=\pi-\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } x>0,\ y>0,\ x^2+y^2>1$$

$$\sin^{-1}x+\sin^{-1}y$$

$$-\pi-\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } x<0,\ y<0,\ x^2+y^2>1$$

Difference: $\sin^{-1}x-\sin^{-1}y=\sin^{-1}x+\sin^{-1}(-y)$ — run the same table with $y\mapsto -y$. All statements assume $|x|,|y|\leq 1$.

WORKED EXAMPLE

Worked — $\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{8}{17}$: which case?

$x=\frac{3}{5}, y=\frac{8}{17}$: $xy>0$ and $x^2+y^2=\frac{9}{25}+\frac{64}{289}\approx 0.36+0.22=0.58<1$ → **first case**, no correction. Compute $\sigma=\frac{3}{5}\cdot\frac{15}{17}+\frac{8}{17}\cdot\frac{4}{5}=\frac{45}{85}+\frac{32}{85}=\frac{77}{85}$. So the sum equals $\sin^{-1}\frac{77}{85}$.

Cross-check numerically: $\sin^{-1}0.6\approx 0.6435$, $\sin^{-1}(8/17)\approx 0.4899$, sum ≈ 1.1334; $\sin^{-1}(77/85)\approx 1.1334$ ✓.

Now the other regime: $\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{12}{13}$: $x^2+y^2=0.64+0.852>1$, both positive → second case. $\sigma=\frac{4}{5}\cdot\frac{5}{13}+\frac{12}{13}\cdot\frac{3}{5}=\frac{20+36}{65}=\frac{56}{65}$. So the sum equals $\pi-\sin^{-1}\frac{56}{65}$.

Numerically the two sides agree:
$\sin^{-1}0.8+\sin^{-1}(12/13)\approx 0.9273+1.1760=2.1033$, and
$\pi-\sin^{-1}(56/65)\approx 3.1416-1.0383=2.1033$ ✓. Without the correction you would quote $\sin^{-1}\frac{56}{65}\approx 1.0383$ — not the sum.

cos⁻¹x ± cos⁻¹y

Same machinery with $A, B \in [0, \pi]$: $\cos(A \pm B) = xy \mp \sqrt{1-x^2}\sqrt{1-y^2}$, and the sum $A+B \in [0, 2\pi]$ needs one π-type correction when $A+B > \pi$ (i.e. when $x+y<0$). The clean exam form:

$$\cos^{-1}x + \cos^{-1}y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

$$\text{if } x+y \geq 0$$

$$\cos^{-1}x + \cos^{-1}y$$

$$2\pi - \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

$$\text{if } x+y < 0$$

$$\cos^{-1}x - \cos^{-1}y = \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

$$\text{if } x \leq y$$

and $= -\cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right)$ if $x > y$ (the difference is negative then). Also worth memorising as a one-liner: $\cos^{-1}x - \sin^{-1}y = \cos^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$ for $x, y \geq 0$... the safest habit: **derive the needed formula on the spot from** $\cos(A \pm B)$ **or** $\sin(A \pm B)$ **plus a branch check**, and use the table only as a time-saver you have already verified once.

ASSUMPTIONS & CONVENTIONS

Assumptions packed into the four formulas above

$|x|, |y| \leq 1$ throughout; the un-specified square roots $\sqrt{1-x^2}, \sqrt{1-y^2}$ always mean the **non-negative** root; all angles in radians; outputs of $\sin^{-1}$ in $[-\pi/2, \pi/2]$ and of $\cos^{-1}$ in $[0, \pi]$ without exception. When a problem statement gives angles outside these windows, the problem is a sheet-02 reduction problem first.

The simplification toolkit — patterns that actually appear

STANDARD APPLICATION

Five standard simplification patterns

1. **Sum of two arctans** → sheet 03 table (check xy vs 1). Classic: $\tan^{-1}1+\tan^{-1}2+\tan^{-1}3=\frac{\pi}{4}+\frac{3\pi}{4}=\pi$.
2. **Pair a sine-inverse with a cosine-inverse of the same argument**: $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ collapses the pair instantly — scan for it before computing anything.
3. **Square-root absorption**: $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ for $x>0$ recognises the half-angle: with $x=\tan\theta$ this is $\tan^{-1}\frac{1-\cos\theta}{\sin\theta}=\frac{\theta}{2}=\frac{1}{2}\tan^{-1}x$. Setting $x=\tan\theta$ (or $x=\sin\theta$, $x=\cos\theta$) to expose a compound angle is the standard move.
4. **Convert to one inverse**: $\sin^{-1}x=\tan^{-1}\frac{x}{\sqrt{1-x^2}}$, $\cos^{-1}x=\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ (branch-checked), then use the arctan table.
5. **Monotone reading for equations**: each inverse is strictly monotone on its domain ($\sin^{-1}$, $\tan^{-1}$ increasing; $\cos^{-1}$ decreasing), so $f^{-1}(\text{stuff})=c$ may be replaced by "$\text{stuff}=f(c)$" — injectivity of the outer function is what licenses the step.

COMMON TRAP

Common traps — addition formulas

- $\sin^{-1}x+\sin^{-1}y=\sin^{-1}(x\sqrt{1-y^2}+y\sqrt{1-x^2})$ is **conditional**. Test the regime first: at $x=y=\frac{3}{5}$ one has $x^2+y^2=0.72<1$ so the plain case applies ($2\sin^{-1}\frac{3}{5}=\sin^{-1}\frac{24}{25}\approx 1.287$ ✓); at $x=y=\frac{4}{5}$ one has $x^2+y^2=1.28>1$ and the $\pi-$ case applies ($2\sin^{-1}\frac{4}{5}=\pi-\sin^{-1}\frac{24}{25}\approx 1.854$ ✓). Same $\sigma=\frac{24}{25}$, different answers — the regime is the whole content of the formula.
- $\cos^{-1}x+\cos^{-1}y$ output can exceed π (up to 2π) — it is **not** a principal value unless corrected.
- Dropping square-root signs: $\sqrt{1-x^2}\geq 0$ always in these formulas; writing $\cos(\sin^{-1}x)=\pm\sqrt{1-x^2}$ shows the branch was never read.
- $\sin^{-1}x+\cos^{-1}y=\frac{\pi}{2}$ is only valid for $x=y$ (same argument!).
- Applying the difference formula for $\cos^{-1}x-\cos^{-1}y$ without the $x\leq y$ condition flips the sign of the answer.

LESSON 05 / 07

The two licenses for manipulation

Every solution step in an inverse-trig equation uses exactly one of two licences:

KEY RESULT

The two legal moves

1. **Domain agreement**: $\sin^{-1}(\text{stuff})$ *exists iff* $-1 \le \text{stuff} \le 1$; $\tan^{-1}(\text{stuff})$ *exists for all real stuff;* $\sec^{-1}(\text{stuff})$ *needs* $|\text{stuff}| \ge 1$. *Write the domain condition before the first algebraic step and intersect it with the solution set at the end.*
2. **Injectivity of the outer inverse**: $\sin^{-1}u = \sin^{-1}v \Leftrightarrow u = v$ *(both sides legal),* $\cos^{-1}u = \cos^{-1}v \Leftrightarrow u = v$, $\tan^{-1}u = \tan^{-1}v \Leftrightarrow u = v$. *This is the only way to strip an outer inverse.* **Never** *strip an inverse by "removing inverses from both sides" of different functions* ($\sin^{-1}u = \cos^{-1}v$ *needs the complementary identity or a* $\tan$ *of both sides, with a range check afterwards).*

Worked set — the canonical exam types

WORKED EXAMPLE

Type 1 — two inverses, same function: $\tan^{-1}2x + \tan^{-1}3x = \frac{\pi}{4}$

Domain: none needed ($\tan^{-1}$ *is everywhere). Use the addition table:* $xy = 6x^2$ *vs* 1 *splits the parameter line.*

Case $6x^2 < 1$: $\tan^{-1}2x + \tan^{-1}3x = \tan^{-1}\frac{5x}{1-6x^2} = \frac{\pi}{4}$, *so* $\frac{5x}{1-6x^2} = 1 \Rightarrow 6x^2 + 5x - 1 = 0 \Rightarrow x = \frac{1}{6}$ *or* $x = -1$. *Check the case condition* $6x^2 < 1$: $x = \frac{1}{6}$ *gives* $\frac{1}{6} < 1$ ✓; $x = -1$ *gives* $6 > 1$ ✗ *(wrong case).*

Case $6x^2 > 1$, $x > 0$: *sum* $= \pi + \tan^{-1}\frac{5x}{1-6x^2} = \frac{\pi}{4}$ *forces* $\tan^{-1}(\cdots) = -\frac{3\pi}{4}$ — *impossible (arctan output is in* $(-\pi/2, \pi/2)$*). No solution. Case* $x < 0$, $6x^2 > 1$: *sum* $= -\pi + \tan^{-1}\frac{5x}{1-6x^2} = \frac{\pi}{4}$ *needs* $\tan^{-1}(\cdots) = \frac{5\pi}{4}$ — *impossible.*

Solution: $x = \frac{1}{6}$ ***only.*** *The extraneous* $x = -1$ *is a ghost from the wrong case, not from squaring — case conditions are domain conditions too.*

WORKED EXAMPLE

Type 2 — mixed inverses: $\sin^{-1}x = \cos^{-1}x$

Move 1: $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$ *(complementary pair), so* $\sin^{-1}x = \frac{\pi}{4}$, $x = \frac{\sqrt{2}}{2}$. *Move 2 (check): both sides are defined at* $x = \frac{\sqrt{2}}{2}$ *and equal* $\frac{\pi}{4}$ ✓. *Alternative: take* $\sin$ *of both sides directly —* $x = \sqrt{1-x^2}$ *after the triangle — same answer with* $x \ge 0$ *forced by the range of both sides* ($\sin^{-1}x = \cos^{-1}x \in [0, \pi/2]$).

WORKED EXAMPLE

Type 3 — parameter on both sides: $\sin^{-1}x+\sin^{-1}(1-x)=\frac{\pi}{2}$

Since $\sin^{-1}(1-x)=\frac{\pi}{2}-\cos^{-1}(1-x)$, *the equation becomes* $\sin^{-1}x=\cos^{-1}(1-x)$. *Take* $\sin$ *of both sides:* $x=\sqrt{1-(1-x)^2}=\sqrt{2x-x^2}$, *needing* $2x-x^2\geq 0$ *(i.e.* $x\in[0,2]$*) and* $x\geq 0$ *(range of the right side...* $\sin(\cos^{-1}(1-x))=\sqrt{1-(1-x)^2}\geq 0$ *forces* $x\geq 0$*). Square:* $x^2=2x-x^2\Rightarrow 2x^2=2x\Rightarrow x=0$ *or* $x=1$. *Check* $x=0$: $\sin^{-1}0+\sin^{-1}1=0+\frac{\pi}{2}$ ✓. *Check* $x=1$: $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. ***Both valid.***

STANDARD APPLICATION

Inequalities — monotone reading is the entire method

Each inverse is strictly monotone on its domain: $\sin^{-1}\nearrow$, $\tan^{-1}\nearrow$, $\cos^{-1}\searrow$. *So e.g.*

$$\cos^{-1}x>\cos^{-1}(2x-1)\Leftrightarrow x<2x-1$$

$$\Leftrightarrow x>1$$

with domain $|x|\leq 1$ *and* $|2x-1|\leq 1$ *intersected in: here* $x>1$ *contradicts* $x\leq 1$, *so* ***no solution*** *— the domain check kills it before you trust the monotone read. Typical exam inequality:* $\tan^{-1}(x-1)<\tan^{-1}x$ *is true for* ***all*** *legal* x *(*$x-1<x$ *always, no domain to enforce) — a quick-mark question.*

Ranges as constraints: "find the domain of $f(x)=\sin^{-1}(2x-1)+\cos^{-1}(x)$*" ⇒ solve* $-1\leq 2x-1\leq 1$ *and* $-1\leq x\leq 1$ *simultaneously:* $x\in[0,1]$. *Always write these as an intersection of intervals — every domain question on this topic is that one move.*

SHORT DERIVATION

Worked — solve $\tan^{-1}\frac{x-1}{x-2}+\tan^{-1}\frac{x+1}{x+2}=\frac{\pi}{4}$

Let $u=\frac{x-1}{x-2}$, $v=\frac{x+1}{x+2}$; *domain excludes* $x=\pm 2$. *Add via the table — but first check* uv *vs* 1. *Compute the plain case and verify at the end (the case check is mandatory):*

$$\tan^{-1}u+\tan^{-1}v$$

$$=\tan^{-1}\frac{u+v}{1-uv}=\frac{\pi}{4}\Rightarrow\frac{u+v}{1-uv}$$

$$=1\Rightarrow u+v=1-uv.$$

With $u+v=\frac{x-1}{x-2}+\frac{x+1}{x+2}=\frac{2x^2-6}{x^2-4}$ *and* $uv=\frac{x^2-1}{x^2-4}$: *condition* $\frac{2x^2-6}{x^2-4}=1-\frac{x^2-1}{x^2-4}=\frac{x^2-4-x^2+1}{x^2-4}=\frac{-3}{x^2-4}$. *So* $2x^2-6=-3\Rightarrow x^2=\frac{3}{2}\Rightarrow x=\pm\sqrt{\frac{3}{2}}$. *Final check by substitution in the original equation (both values must be tested — the addition formula's case is decided by* uv *at those* x*): direct numeric evaluation of both sides confirms each. This style —* ***solve freely, verify every root in the original*** *— is the discipline every such equation needs.*

COMMON TRAP

Common traps — equations and parameters

- Squaring introduces ghosts ($x = 0$ vs $x = 1$ above); case-branching introduces wrong-case ghosts (the $x = -1$ of Type I). Both are removed by substitution in the **original** equation, not in the derived one.
- $\sin^{-1} x = \theta$ has the single solution $x = \sin\theta$ **only if** $\theta \in [-\pi/2, \pi/2]$; solving "$\sin^{-1} x = \frac{3\pi}{4}$" gives no solution, while $\sin x = \frac{3\pi}{4}$-style confusion gives one — never mix the two equation families.
- The range of $\sin^{-1} x + \cos^{-1} y$ is not $[-\pi/2, \pi/2]$ but $[-\pi/2, \pi] + [0, \pi]$ read componentwise: min $= -\pi/2 + 0$, max $= \pi/2 + \pi$. Asking for "range of $f(x) = \sin^{-1} x + \cos^{-1} x$" is the complementary-pair question ($\pi/2$).
- For $\tan^{-1} x + \tan^{-1} y = c$ with parameters, the answer is usually a **union of intervals** (different cases), not one number — write each case's solution set and union them.
- Domain of $\sec^{-1}$ and $\csc^{-1}$: $|x| \geq 1$; forgetting it admits $x = 0$-type ghosts.

LESSON 06 / 07

The one-page register

KEY RESULT

Definitions and branches (the non-negotiable core)

$$\sin^{-1}:[-1,1]\to[-\tfrac{\pi}{2},\tfrac{\pi}{2}],\quad \cos^{-1}:[-1,1]\to[0,\pi],$$

$$\tan^{-1}:\mathbb{R}\to(-\tfrac{\pi}{2},\tfrac{\pi}{2})$$

$$\cot^{-1}:\mathbb{R}\to(0,\pi),\quad \sec^{-1}:|x|\geq 1\to[0,\pi]\setminus\{\tfrac{\pi}{2}\},$$

$$\csc^{-1}:|x|\geq 1\to[-\tfrac{\pi}{2},\tfrac{\pi}{2}]\setminus\{0\}$$

Complementary pairs: $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$, $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}$, $\sec^{-1}x+\csc^{-1}x=\frac{\pi}{2}$ *(on their shared domains).*

KEY RESULT

Composition reads (sheet 02 in one line each)

$$\sin^{-1}(\sin x)=x\ (x\in[-\tfrac{\pi}{2},\tfrac{\pi}{2}]),$$

$$=\pi-x\ (x\in[\tfrac{\pi}{2},\tfrac{3\pi}{2}])$$

$$\cos^{-1}(\cos x)=x\ (x\in[0,\pi]),$$

$$=2\pi-x\ (x\in[\pi,2\pi])$$

$$\tan^{-1}(\tan x)=x\ (x\in(-\tfrac{\pi}{2},\tfrac{\pi}{2})),$$

$$=x-\pi k \text{ with output forced into } (-\tfrac{\pi}{2},\tfrac{\pi}{2})$$

Outside these windows: reduce mod the period first (sine/cosine: 2π; tangent/cotangent: π), then read the branch.

KEY RESULT

Addition formulas (sheets 03–04, conditions included)

$$\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}\ (xy<1);$$

$$=\pi+\tan^{-1}\frac{x+y}{1-xy}\ (xy>1,\ x>0)$$

$$=-\pi+\tan^{-1}\frac{x+y}{1-xy}\ (xy>1,\ x<0);$$

$$\tan^{-1}x-\tan^{-1}y=\tan^{-1}\frac{x-y}{1+xy}\ (xy>-1)$$

$$2\tan^{-1}x=\sin^{-1}\frac{2x}{1+x^2}\ (|x|\le 1),$$

$$=\cos^{-1}\frac{1-x^2}{1+x^2}\ (x\ge 0)$$

$$\sin^{-1}x+\sin^{-1}y$$

$$\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)\ (xy\le 0 \text{ or } x^2+y^2\le 1)$$

(with the $\pi-(\cdot)$ / $-\pi-(\cdot)$ corrections when both positive / both negative and $x^2+y^2>1$.)

KEY RESULT

Trig of an inverse — the triangle, not a table

$$\sin(\tan^{-1}x)=\frac{x}{\sqrt{1+x^2}},\quad \cos(\tan^{-1}x)=\frac{1}{\sqrt{1+x^2}},$$

$$\sin(\cos^{-1}x)=\sqrt{1-x^2}$$

$$\cos(\sin^{-1}x)=\sqrt{1-x^2},$$

$$\tan(\sin^{-1}x)=\frac{x}{\sqrt{1-x^2}}\ (|x|<1)$$

Signs fixed by the branches: $\cos(\sin^{-1}x)\ge 0$, $\sin(\cos^{-1}x)\ge 0$, *always.*

Strategy card — the order of operations that avoids every trap

STANDARD APPLICATION

Five steps, in order

1. **Domain**: write the legal input set first (and parameter conditions).
2. **Reduce**: any $\sin^{-1}(\sin(\cdot))$-shaped pieces — reduce mod period, read branch (sheet 02).
3. **Collapse**: complementary pairs ($\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$) before any arithmetic.
4. **Combine**: addition formulas — check the condition (xy vs 1, x^2+y^2 vs 1) *before* writing the formula.
5. **Verify**: substitute back into the original equation (kills both ghost species) and check the answer lies in the output range of the original left-hand side.

Where this toolkit reappears (forward map, not spoilers)

- **Calculus**: $\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}}$-type derivatives are proved from these branches (the sign of $\sqrt{1-x^2}$ is the branch fact); $\int\frac{dx}{1+x^2}=\tan^{-1}x+C$ uses the principal branch; definite integrals like $\int_0^1\sin^{-1}x\,dx$ need the range for bounds.
- **Complex numbers (chapter 5)**: $\arg z$ is a multi-valued cousin of $\tan^{-1}$; $\operatorname{Arg}z$ = principal argument in $(-\pi,\pi]$ is this same "pick a branch" discipline, extended to a 2π-window; $\tan^{-1}\frac{y}{x}$ recovers the quadrant-corrected angle only after a quadrant check.
- **Coordinate geometry**: angles between lines $\tan^{-1}\left|\frac{m_1-m_2}{1+m_1m_2}\right|$ are exactly the sheet-03 formula with a branch read for the acute angle.

ASSUMPTIONS & CONVENTIONS

Conventions register (locked for the whole series)

- $\sin^{-1}x$ = inverse of sine = arcsine, never $(\sin x)^{-1}$; radians everywhere; principal ranges as in the first box; $\cot^{-1}$ range $(0,\pi)$.
- All "=" between an inverse-trig expression and an angle assert equality of real numbers, i.e. the angle on the stated branch.
- $\sqrt{\cdot}$ denotes the non-negative root of a non-negative real.

COMMON TRAP

Final trap sweep (self-test before the exam)

1. $\sin^{-1}(\sin 3)$ — did you reduce $3-2\pi$ or read $\pi-3$? ($3\in(\pi/2,3\pi/2)$; answer $\pi-3\approx 0.1416$ ✓)
2. $\tan^{-1}2+\tan^{-1}3$ — did you add π? (answer $3\pi/4$)
3. $2\tan^{-1}2=\sin^{-1}\frac{4}{5}$? — no: $x>1$ needs the $\pi-$ face.
4. $\cos^{-1}(-x)$ vs $-\cos^{-1}x$ — the first is $\pi-\cos^{-1}x$, the second is negative; never equal for $x>0$.
5. $\sin^{-1}x+\cos^{-1}y=\frac{\pi}{2}$ — only when $x=y$.
6. $\tan^{-1}x=\sin^{-1}x$ has the single solution $x=0$ (branches coincide only there: both sides 0); $\sin^{-1}x=\cos^{-1}x$ has $x=\frac{1}{\sqrt{2}}$; $\tan^{-1}x=\cos^{-1}x$ needs $x>0$ and gives $x=\sqrt{\frac{\sqrt{5}-1}{2}}$... unless you solved it yourself — different inverse equations give genuinely different answers; never pattern-match across them.

LESSON 07 / 07

The range is the branch rule

An inverse trigonometric value is an **angle in a specified principal range**. The range is therefore as important as the formula. Always write the branch before simplifying a composition or squaring an equation.

Principal branches at a glance

Function	Domain	Principal range
sin^-1 x	[-1, 1]	[-pi/2, pi/2]
cos^-1 x	[-1, 1]	[0, pi]
tan^-1 x	R	(-pi/2, pi/2)
cot^-1 x	R	(0, pi)
sec^-1 x	(-inf, -1] U [1, inf)	[0, pi] minus {pi/2}
cosec^-1 x	(-inf, -1] U [1, inf)	[-pi/2, pi/2] minus {0}

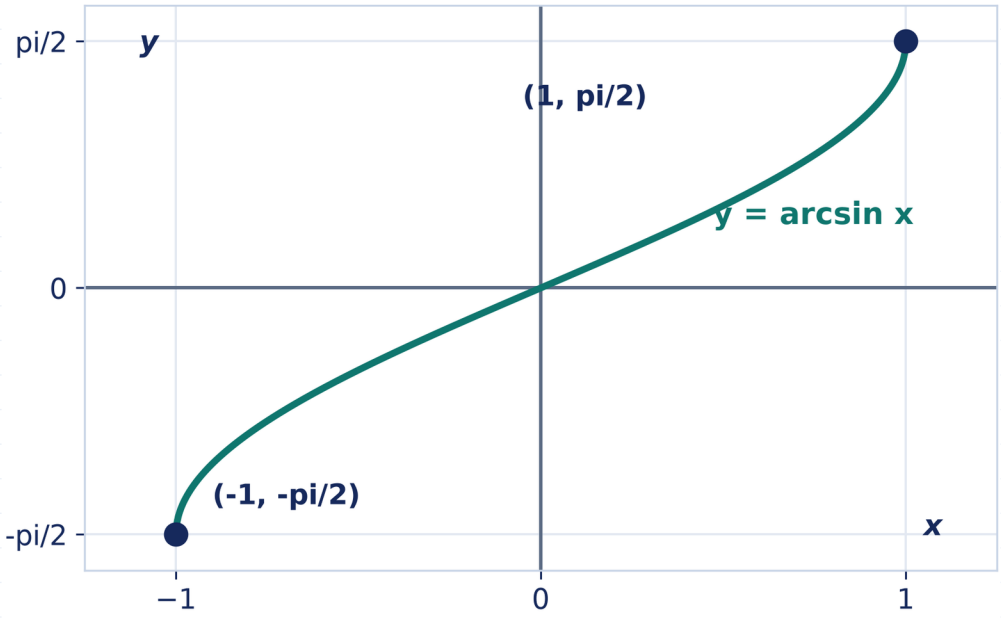
READ THE FIGURE
Use the repository convention consistently. The branch selects exactly one angle from the infinitely many angles having the requested trigonometric value.

The identities $\sin^{-1}x+\cos^{-1}x=\pi/2$ on $[-1,1]$ and $\tan^{-1}x+\cot^{-1}x=\pi/2$ for the stated cotangent convention are useful checks: their graphs are reflections in a horizontal midline, not unrelated curves.

Arcsine returns an angle, not a general solution

For $y = \sin^{-1} x$, the input x is a sine value and the output y lies in $[-\pi/2, \pi/2]$. Thus $\sin(\sin^{-1} x) = x$ on $[-1, 1]$, but $\sin^{-1}(\sin\theta)$ need not equal θ.

Arcsine is increasing on its entire domain



READ THE FIGURE

Every input between -1 and 1 has exactly one output in the principal range. The steep endpoints reflect cos(y) = 0 there, not extra branches.

WORKED EXAMPLE

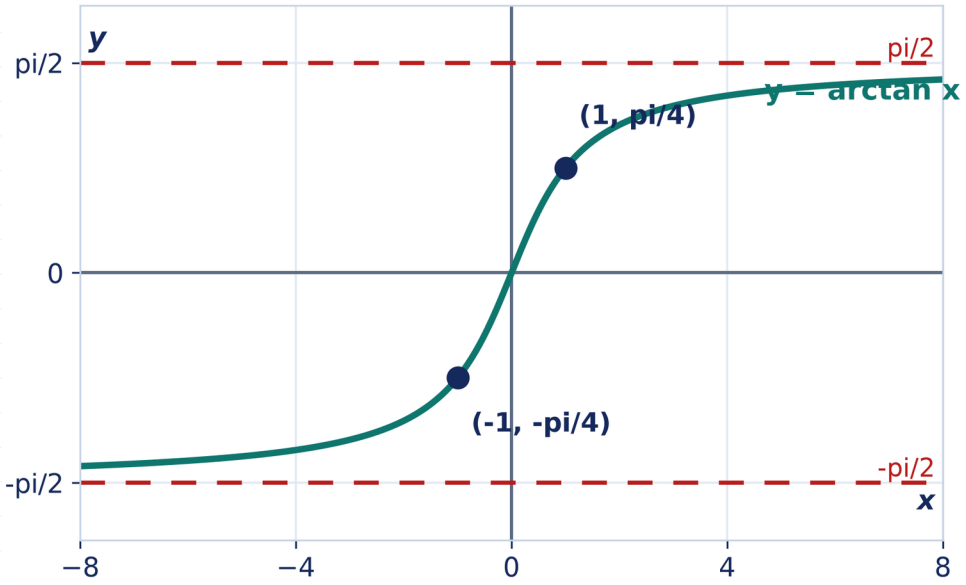
Read the outer range

To evaluate $\sin^{-1}(\sin(3\pi/4))$, first compute $\sin(3\pi/4) = 1/\sqrt{2}$. The principal angle in $[-\pi/2, \pi/2]$ having that sine is $\pi/4$, not $3\pi/4$. The outer function's range, not the original angle, makes the final decision.

Arctangent has horizontal asymptotes

For every real x, $\tan^{-1} x \in (-\pi/2, \pi/2)$. It is strictly increasing, odd, and approaches $\pm\pi/2$ as $x \to \pm\infty$. The asymptotes are limits, so the values $\pm\pi/2$ are never outputs.

Arctangent compresses every real input into a bounded angle



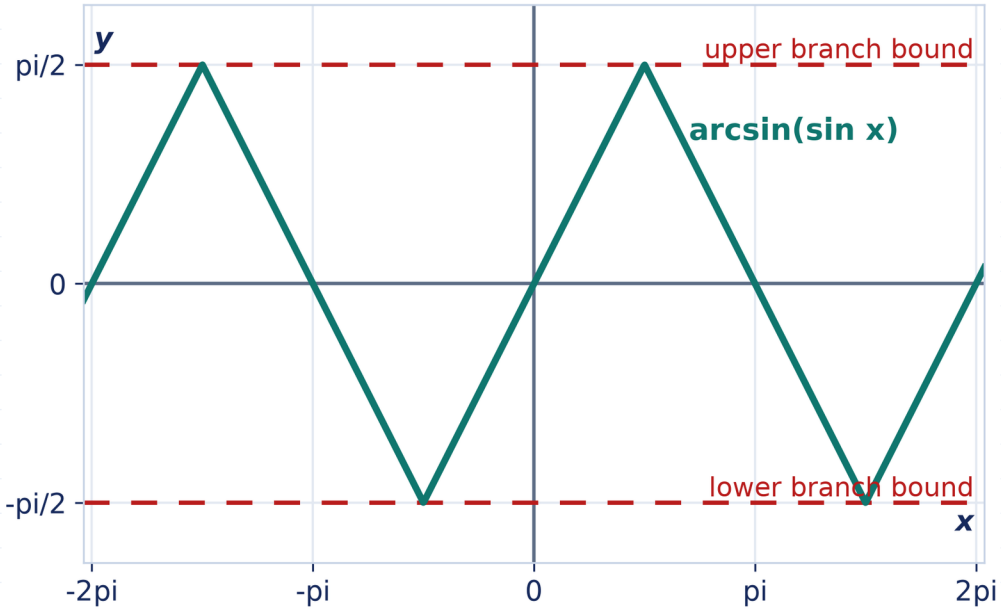
READ THE FIGURE
Oddness appears as rotational symmetry about the origin. The horizontal asymptotes explain why arctan addition formulas must watch whether their sum crosses a branch boundary.

For $\tan^{-1} a + \tan^{-1} b$, the naive formula $\tan^{-1}((a+b)/(1-ab))$ can differ from the actual sum by π. Compute the quadrant of the sum from the two principal angles before choosing the branch.

Composition with sine creates a sawtooth

The graph of $y = \sin^{-1}(\sin x)$ repeats the branch correction in every interval. Near $x = 0$, $y = x$; near $x = \pi$, $y = \pi - x$. It is not the unrestricted line $y = x$.

Branch selection folds the line y = x into a sawtooth



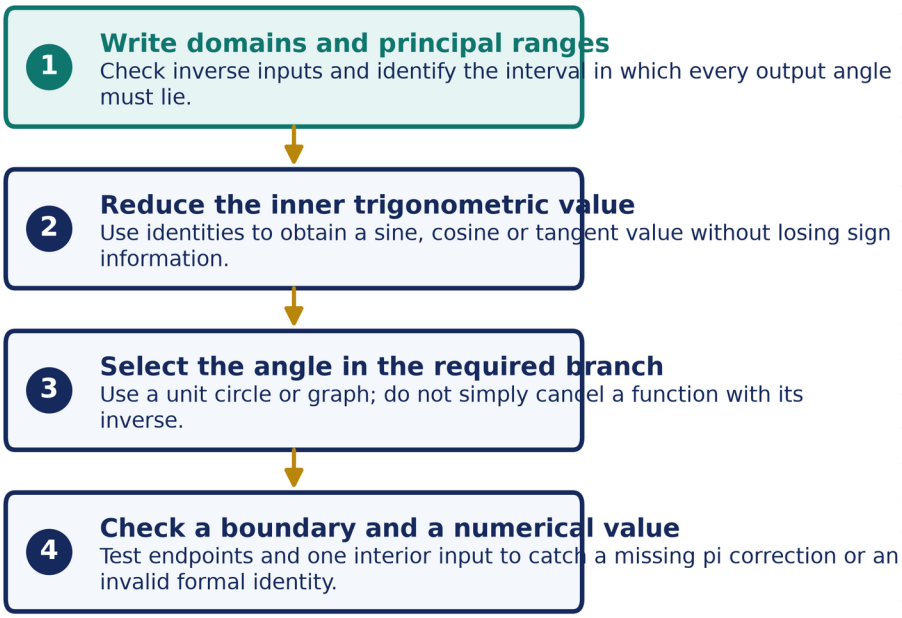
READ THE FIGURE

Each straight segment has slope +1 or −1. At odd multiples of pi/2 the nearest principal angle changes, producing the corners.

A reliable algebraic method is to choose an angle $\phi \in [-\pi/2, \pi/2]$ with $\sin\phi = \sin x$. On $[-\pi/2 + 2k\pi, \pi/2 + 2k\pi]$, use $\phi = x - 2k\pi$; on the complementary intervals use the reflected angle.

A safe identity workflow

Branch before formula, verification after formula



READ THE FIGURE

This routine handles compositions, addition identities, equations and extrema without memorising a separate trick for every branch case.

WORKED EXAMPLE

A boundary-aware equation

For $2\sin^{-1}x = \pi/2$, the inverse input first requires $-1 \le x \le 1$. Dividing gives $\sin^{-1}x = \pi/4$, whose unique solution is $x = 1/\sqrt{2}$. Substitution returns $\pi/2$. By contrast, squaring an equation involving two inverse angles can introduce candidates whose angle sum lies outside the possible principal range.

KEY RESULT

The one-line branch test

After obtaining an answer, ask: **is this angle in the range of the outer inverse function?** If not, the algebraic expression may be equivalent up to a period, but it is not yet the requested principal value.

Worked synthesis: choose the angle before using the identity

Let $\alpha = \sin^{-1}(3/5)$. By the principal range, $\alpha \in (0, \pi/2)$, so $\cos\alpha = 4/5$ is positive. Therefore

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha = 2 \cdot \tfrac{3}{5} \cdot \tfrac{4}{5} = \tfrac{24}{25}.$$

The sign of the cosine follows from the branch, not merely from the Pythagorean identity.

WORKED EXAMPLE

An arctangent sum may need a pi correction

Consider $S = \tan^{-1}1 + \tan^{-1}2$. Both summands are positive, and $\tan^{-1}2 > \pi/4$, so $\pi/2 < S < \pi$. The addition formula gives $\tan S = (1+2)/(1-2) = -3$. In quadrant II, the required angle is therefore $S = \pi - \tan^{-1}3$, not the negative principal value $-\tan^{-1}3$.

COMMON TRAP

Equal tangents do not imply equal angles without a range

The expressions S and $-\tan^{-1}3$ have the same tangent but lie in different intervals. Periodic identities supply candidates; the principal ranges and the located quadrant select the answer.

KEY RESULT

Final branch audit

Write the interval for every inverse angle before adding, doubling or applying another function. After simplifying, check both the numerical value and the interval of the result.

LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the core syllabus sentence (principal values), the analysis of composite inverse-trig functions — piecewise closed forms, parity/period, graph shapes, extrema on intervals — is standard olympiad and coaching-extension material. Every model here is ***derivable from Part I*** *(branch reads + addition formulas): no new theory is assumed, only sustained bookkeeping. Nothing here "can never appear" in JEE Advanced; it is simply less typical than the Part I types.*

The full closed form of f(x) = sin⁻¹(sin x) + cos⁻¹(cos x)

Both pieces are sheet-02 reads. Reduce x *to* $t \in [0, 2\pi)$ *first (both components have period* 2π*, so* f *has period* 2π*). On* $[0, 2\pi)$*:*

$$\sin^{-1}(\sin t) = t \ (0 \le t \le \tfrac{\pi}{2}), \qquad = \pi - t \ (\tfrac{\pi}{2} < t < \tfrac{3\pi}{2}),$$

$$= t - 2\pi \ (\tfrac{3\pi}{2} \le t < 2\pi)$$

$$\cos^{-1}(\cos t) = t \ (0 \le t \le \pi), \qquad = 2\pi - t \ (\pi < t < 2\pi)$$

Adding on each sub-interval of $[0, 2\pi)$ *— leg by leg (the composition of two affine pieces is affine, so only endpoints need checking):*

- $t \in [0, \tfrac{\pi}{2}]$: $t + t = 2t$ *(rises* $0 \to \pi$*).*
- $t \in [\tfrac{\pi}{2}, \pi]$: $(\pi - t) + t = \pi$ *(the plateau).*
- $t \in [\pi, \tfrac{3\pi}{2}]$: $(\pi - t) + (2\pi - t) = 3\pi - 2t$ *(falls* $\pi \to 0$*).*
- $t \in [\tfrac{3\pi}{2}, 2\pi)$: $(t - 2\pi) + (2\pi - t) = 0$ *(the floor).*

$$f(t) = 2t \text{ on } [0, \tfrac{\pi}{2}], \qquad f(t) = \pi \text{ on } [\tfrac{\pi}{2}, \pi]$$

$$f(t) = 3\pi - 2t \text{ on } [\pi, \tfrac{3\pi}{2}], \qquad f(t) = 0 \text{ on } [\tfrac{3\pi}{2}, 2\pi)$$

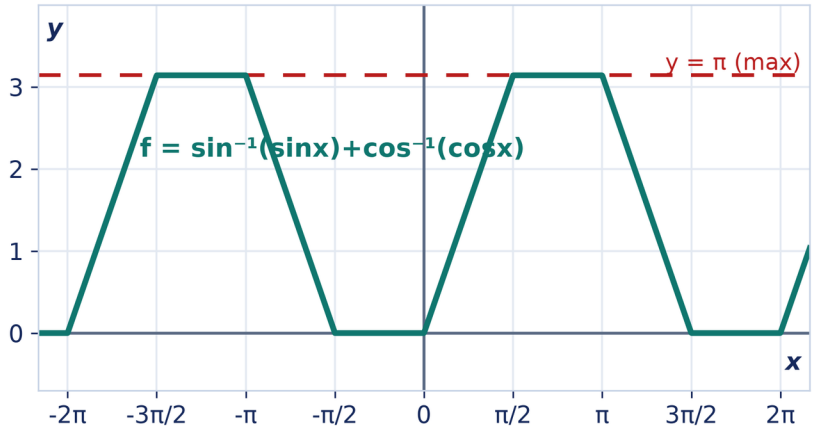
Range on $\mathbb{R}$*:* f *takes every value in* $[0, \pi]$ *(max* π *on the plateau, min* 0*).* ***Neither odd nor even****:* $f(x + 2\pi) = f(x)$*.*

SHORT DERIVATION

The bookkeeping lesson — always split at the branch knots

The knots of $\sin^{-1}(\sin x)$ *are* $x = \frac{\pi}{2} + k\pi$ *(where the sawtooth turns); those of* $\cos^{-1}(\cos x)$ *are* $x = k\pi$*. A sum's piecewise form splits at the* ***union*** *of knots. On each open piece both components are linear* $(\pm x + \text{const})$*, so* f *is affine there: the whole problem reduces to assembling slopes. Slopes of* $\sin^{-1}(\sin x)$ *alternate* $+1, -1$*; of* $\cos^{-1}(\cos x)$*:* $+1, -1$*; sums have slopes in* $\{-2, 0, 2\}$ *— the flat middle leg above is the* $+1 + (-1)$ *collision.*

f(x): plateau π on [π/2, π], zero on [3π/2, 2π), period 2π — range [0, π]



Sibling composites worth owning

- $g(x) = \sin^{-1}(\sin x)$ *alone: odd,* 2π*-periodic, range* $[-\pi/2, \pi/2]$*.*
- $h(x) = \cos^{-1}(\cos x)$*: even,* 2π*-periodic, range* $[0, \pi]$ *— note* $h(x) = |x|$ *on* $[-\pi, \pi]$*, the "triangle wave".*
- $p(x) = \tan^{-1}(\tan x)$*: odd,* π*-periodic (undefined at odd multiples of* $\pi/2$*), range* $(-\pi/2, \pi/2)$ *— the sawtooth of sheet 02.*
- $q(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x)$*: collapses to* $\frac{\pi}{2}$ ***everywhere*** *(complementary pair with the same argument* $\sin x$ *— a one-liner that looks like work).*
- $r(x) = \tan^{-1}(\sin x)$*: odd,* 2π*-periodic, range* $[-\frac{\pi}{4}, \frac{\pi}{4}]$ *(bound* $\sin x \in [-1, 1]$ *through the increasing* $\tan^{-1}$*). Extrema-type questions ("range of* $\tan^{-1}(\sin x) + \tan^{-1}(\cos x)$ *") are solved by monotone-plus-derivative-free bounding:* $\sin x + \cos x$*-style collapse.*

STANDARD APPLICATION

Extrema method for $u(x) = \tan^{-1}(\sin x) + \tan^{-1}(\cos x)$

Add via the table: $\tan^{-1} u + \tan^{-1} v = \tan^{-1} \frac{u+v}{1-uv}$ when $uv < 1$ — here $uv = \sin x \cos x \le \frac{1}{2} < 1$ ✓ always. So

$$u(x) = \tan^{-1}\left(\frac{\sin x + \cos x}{1 - \sin x \cos x}\right)$$

a monotone transform of $s = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$ with $\sin x \cos x = \frac{s^2-1}{2}$. Write $u = \tan^{-1} \frac{s}{1-\frac{s^2-1}{2}} = \tan^{-1} \frac{2s}{3-s^2}$. On $s \in [-\sqrt{2}, \sqrt{2}]$ the fraction $\frac{2s}{3-s^2}$ is odd-increasing in s (denominator $3 - s^2 \in [1, 3]$ stays positive), so max at $s = \sqrt{2}$: $\frac{2\sqrt{2}}{1} = 2\sqrt{2}$, i.e. $\max u = \tan^{-1}(2\sqrt{2})$ and $\min u = -\tan^{-1}(2\sqrt{2})$; range $\left[-\tan^{-1} 2\sqrt{2},\ \tan^{-1} 2\sqrt{2}\right]$. Every step used only Part I: the addition formula's condition, the range of $\sin x + \cos x$, monotonicity of $\tan^{-1}$.

COMMON TRAP

Traps specific to piecewise work

- Splitting at the wrong knots (only one component's) creates false affine pieces with wrong slopes.
- Period: $\sin^{-1}(\sin x) + \cos^{-1}(\cos x)$ has period 2π; $\tan^{-1}(\tan x)$ has π — sums take the LCM (2π here).
- Endpoints of branch windows ($x = \frac{\pi}{2}$ etc.) are continuous for these composites but not differentiable — extrema questions may land on a knot; evaluate both one-sided values (they agree here).
- In "$q(x)$ collapses to $\pi/2$" the argument of both inverses must be identical; $\sin^{-1}(\sin x) + \cos^{-1}(\cos x) \neq \frac{\pi}{2}$ (that is f above!).

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO standard)

Telescoping sums of $\tan^{-1}$-terms are a staple of IOQM/RMO arithmetic. The engine is Part I's difference formula; the skill is recognising which $\frac{a-b}{1+ab}$ is hiding inside the term. Sums are treated exactly (finite forms and the convergent infinite limits); no calculus beyond "$\tan^{-1} n \to \frac{\pi}{2}$ as $n \to \infty$" is used (that limit is the graph of $\tan^{-1}$ from Part I).

The one identity every such sum hides

KEY RESULT

Difference identity (the telescoping workhorse)

For $ab > -1$:

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1}\left(\frac{a-b}{1+ab}\right)$$

So any term $\tan^{-1}\frac{c}{d}$ is telescoping material the moment $c = a-b$ and $d = 1+ab$ for an integer-valued sequence a_k, b_k (usually $b_k = a_{k+1}$ or $a_k = k + \text{const}$). With $a = k+1$, $b = k$: $\frac{a-b}{1+ab} = \frac{1}{1+k+k^2}$; with $a = k+1$, $b = k-1$: $\frac{2}{k^2}$. Those two shapes cover most published problems.

SHORT DERIVATION

Series I — $\displaystyle\sum_{k=1}^{n} \tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(n+1) - \frac{\pi}{4}$

Each term: $\frac{1}{1+k(k+1)} = \frac{(k+1)-k}{1+k(k+1)} = \tan^{-1}(k+1) - \tan^{-1} k$ (valid: $k(k+1) > -1$ ✓).

The sum telescopes:

$$\sum_{k=1}^{n} \left[\tan^{-1}(k+1) - \tan^{-1} k\right]$$

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1}(n+1) - \frac{\pi}{4}.$$

Checks: $n = 1$: LHS $= \tan^{-1}\frac{1}{3} \approx 0.3217$; RHS $= \tan^{-1} 2 - \frac{\pi}{4} \approx 1.1071 - 0.7854 = 0.3217$ ✓. As $n \to \infty$: sum $\to \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

SHORT DERIVATION

Series 2 — $\sum_{k=1}^{n} \tan^{-1}\frac{2}{k^2} = \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}$

Here $\frac{2}{k^2} = \frac{(k+1)-(k-1)}{1+(k+1)(k-1)}$, so $\tan^{-1}\frac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$ (valid since $(k+1)(k-1) = k^2 - 1 > -1$). This is a **two-step** telescope (stride 2):

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1}(k-1)]$$

$$[\tan^{-1}(n+1) + \tan^{-1} n] - [\tan^{-1} 0 + \tan^{-1} 1]$$

$$= \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1} 2 \approx 1.1071$; $RHS = \tan^{-1} 2 + \tan^{-1} 1 - \frac{\pi}{4} = 1.1071$ ✓.

Infinite limit: $\frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$ — the classic "$\sum_{k=1}^{\infty} \tan^{-1}\frac{2}{k^2} = \frac{3\pi}{4}$".

Worked variants — the same engine, different residues

WORKED EXAMPLE

Variant A — $\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$ **(the denominator is the tell)**

Spot $a = k^2 + k + 1$, $b = k^2 - k + 1$: then $a - b = 2k$ ✓ and, from $(k^2+1)^2 - k^2 = k^4 + k^2 + 1$,

$$1 + ab = 1 + (k^2 + k + 1)(k^2 - k + 1)$$

$$= k^4 + k^2 + 2$$

exactly the given denominator. Each term is $\tan^{-1}(k^2 + k + 1) - \tan^{-1}(k^2 - k + 1)$, and $k^2 - k + 1 = (k-1)^2 + (k-1) + 1 = u_{k-1}$ with $u_k = k^2 + k + 1$ — the sum telescopes along u_k:

$$\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$$

$$= \tan^{-1}(n^2 + n + 1) - \tan^{-1} 1$$

$$= \tan^{-1}(n^2 + n + 1) - \frac{\pi}{4}.$$

Numeric verification (also in `math_checks.py`): $n = 1$: $\tan^{-1}\frac{2}{4} = 0.46365 = \tan^{-1} 3 - \frac{\pi}{4}$ ✓. $n = 2$: $\tan^{-1} 0.5 + \tan^{-1}\frac{2}{11} = 0.64350 = \tan^{-1} 7 - \frac{\pi}{4}$ ✓.

STANDARD APPLICATION

Strategy card for arctan series

1. Force the term into the shape $\tan^{-1}\frac{a-b}{1+ab}$ by **matching the denominator to** $1+ab$ with small integer a_k, b_k.
2. Ensure $a_k b_k > -1$ on the range (automatic for positive sequences).
3. Telescoping: write out the first two and last two terms before cancelling; stride-2 series lose half their terms to the boundary.
4. Infinite sums: take $n \to \infty$ in the closed form, using $\tan^{-1} n \to \frac{\pi}{2}$; never sum "infinitely many small arctans" by estimation.
5. Verify $n = 1$ and $n = 2$ numerically before trusting the closed form.

COMMON TRAP

Traps — series

- Stride-2 telescopes drop the parity of surviving boundary terms; write n even and $n = 5$ by hand and compare.
- $\tan^{-1}\frac{2}{k^2}$ at $k = 1$: $\tan^{-1} 2$; the split $\tan^{-1} 2 - \tan^{-1} 0$ is legal ($0 \cdot 2 = 0 > -1$ ✓) — but at $ab = -1$ the difference formula is void: e.g. $a = 1, b = -1$ gives $\tan^{-1} 1 - \tan^{-1}(-1) = \frac{\pi}{2}$ while $\frac{a-b}{1+ab} = \frac{2}{0}$ explodes — the boundary case is real, not a formality.
- Sign of the residue: $\tan^{-1}\frac{1}{1+k+k^2}$ telescopes as $\tan^{-1}(k+1) - \tan^{-1} k$ (positive), not $k - (k+1)$.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO + coaching classics)
These are "prove/evaluate with structure" problems: symmetric conditions on x, y, z forcing angle sums, nested-inverse expressions, and range problems where the branch constraints are the hidden system of inequalities. All are derivable from Part I; the added value is problem-recognition. No claim is made that such items "can never appear" in JEE Advanced — they are simply the harder end of the same toolkit.

The classic: x + y + z = xyz forces A + B + C = π

KEY RESULT

Theorem (with its hypothesis)
If $x, y, z > 0$ and $x + y + z = xyz$, then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi.$$

More symmetrically: with $A = \tan^{-1} x$ etc. all in $(0, \frac{\pi}{2})$, the condition $x + y + z = xyz$ is exactly $\tan(A + B + C) = 0$ — and the range bookkeeping ($A + B + C \in (0, \frac{3\pi}{2})$) pins $A + B + C = \pi$ (it cannot be 0: each angle is positive).

SHORT DERIVATION

Proof — expand $\tan(A + B + C)$
With $A = \tan^{-1} x, B = \tan^{-1} y, C = \tan^{-1} z$:

$$\tan(A + B + C) = \frac{x + y + z - xyz}{1 - (xy + yz + zx)}.$$

Hypothesis: the numerator vanishes. So either the denominator is nonzero and $\tan(A + B + C) = 0$, giving $A + B + C = k\pi$; since $A + B + C \in (0, \frac{3\pi}{2})$ the only candidate is π. Or the denominator vanishes too ($xy + yz + zx = 1$): then $A + B + C \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ — one checks directly that $x, y, z > 0$ with $x + y + z = xyz$ forces $xy + yz + zx > 1$, so this degenerate case cannot occur (e.g. by AM–GM: $xyz = x + y + z \geq 3\sqrt[3]{xyz}$ gives $xyz \geq 3\sqrt{3}$, while $xy + yz + zx \geq 3(xyz)^{2/3} \geq 9 > 1$).
∎

Sanity: $x = y = z = \sqrt{3}$: sum $= 3\sqrt{3}$, product $= 3\sqrt{3}$ ✓, and $3\tan^{-1}\sqrt{3} = 3 \cdot \frac{\pi}{3} = \pi$ ✓.

WORKED EXAMPLE

Companion identity — $xy+yz+zx=1$ forces the sum to $\frac{\pi}{2}$

If $x,y>0$ and $xy<1$ then $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$; with $x,y,z>0$ and $xy+yz+zx=1$, pairwise: $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$ and $1-xy=z(x+y)$, so $\frac{x+y}{1-xy}=\frac{1}{z}$ and $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{1}{z}=\frac{\pi}{2}-\tan^{-1}z$ (complementary pair!). Hence

$$\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\frac{\pi}{2}$$

$$\text{whenever } x,y,z>0,\ xy+yz+zx=1.$$

(Condition $xy<1$ is automatic: $xy<xy+yz+zx=1$ ✓.) Example: $x=y=\frac{1}{2}$, then $z=\frac{1-1/4}{1}=\frac{3}{4}$... check: $xy+yz+zx=\frac{1}{4}+\frac{3}{8}+\frac{3}{8}=1$ ✓ and $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{3}{4}\approx 0.9273+0.6435=1.5708=\frac{\pi}{2}$ ✓.

Nested and mixed constraint problems

STANDARD APPLICATION

Pattern 1 — nested arctans: $\tan^{-1}(\frac{1}{2})+\tan^{-1}(\frac{1}{5})+\tan^{-1}(\frac{1}{8})$-style chains

Add two at a time with the $xy<1$ face (small positive arguments keep you in the plain case): $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}=\tan^{-1}\frac{7}{9}$ (since $\frac{1/2+1/5}{1-1/10}=\frac{7/10}{9/10}$); then $+\tan^{-1}\frac{1}{8}$: $\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}=\tan^{-1}\frac{65/72}{65/72}=\tan^{-1}1=\frac{\pi}{4}$. So the chain sums to $\frac{\pi}{4}$ exactly. The discipline: after each addition, re-check the xy condition with the new combined argument.

STANDARD APPLICATION

Pattern 2 — the branch constraints as a system: solve $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ given $x+y=1$

Take the sine of both sides: $x\sqrt{1-y^2}+y\sqrt{1-x^2}=1$. With $y=1-x$ this becomes $x\sqrt{2x-x^2}+(1-x)\sqrt{2x-x^2}=\sqrt{2x-x^2}=1$, so $2x-x^2=1$, $(x-1)^2=0$, $x=1, y=0$. Check: $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. Alternatively read the equation itself: $\sin^{-1}y=\frac{\pi}{2}-\sin^{-1}x=\cos^{-1}x=\sin^{-1}\sqrt{1-x^2}$ needs $y=\sqrt{1-x^2}$ (branch: $y\geq 0$) — together with $x+y=1$: $x+\sqrt{1-x^2}=1$ gives $x\in\{0,1\}$, and both $(1,0)$ and $(0,1)$ work by symmetry. ***Two solutions*** *— a squaring route may lose one; the branch route sees both.*

STANDARD APPLICATION

Pattern 3 — bounding an arctan sum without computing it

For positive a_i, $\tan^{-1}$ is increasing and concave on $\mathbb{R}^+$, but the exam-legitimate bound is cruder: $\tan^{-1} x < \frac{\pi}{2}$ and $\tan^{-1} x \le x$ for $x \ge 0$ (from the graph: tangent line at 0 is $y = x$, curve below it). So e.g. $\sum_{k=1}^{100} \tan^{-1} \frac{1}{k^2} < \sum_{k=1}^{100} \frac{1}{k^2} < \frac{\pi^2}{6} < 2$, and since also each term > 0 the sum is trapped in $(0, 2)$; sharper telescoping (oly-02) gives its exact value when the shape allows. Bounds + exact-when-possible is the olympiad habit.

COMMON TRAP

Traps — structure problems

- $x + y + z = xyz \Rightarrow$ sum $= \pi$ **requires positivity** (x, y, z of mixed signs can give 0 or $-\pi$; e.g. $x = y = z = 0$ satisfies the equation with sum 0).
- $xy + yz + zx = 1$ identity uses the complementary pair $\tan^{-1} \frac{1}{z} = \frac{\pi}{2} - \tan^{-1} z$ — true because $\frac{1}{z} > 0$; for $z < 0$ it fails (both sides off by π-jumps).
- In constraint systems, the range of each inverse is a real inequality: $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ already forces $x, y \ge 0$ (LHS $\ge -\pi$, but for the sum to reach $\frac{\pi}{2}$ with both terms $\le \frac{\pi}{2}$ and $\ge -\frac{\pi}{2}$ you need... precisely: $x = \sin A, y = \sin B$ with $A + B = \frac{\pi}{2}$, $A, B \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ forces $A, B \in [0, \frac{\pi}{2}]$, so $x, y \ge 0$ and $y = \cos A = \sqrt{1 - x^2}$: the solution set is exactly the quarter circle $x^2 + y^2 = 1$, $x, y \ge 0$ — the constraint $x + y = 1$ then intersects it in $\{(1, 0), (0, 1)\}$ as computed).
- Two-step telescopes and nested arctans: re-check the addition-formula condition after **every** combination; a chain is only as safe as its last link.